Spotify

Spotify Wins Listening, But Struggles With Discovery.

Despite industry-leading recommendation systems, meaningful music discovery remains a **key growth challenge.**



Spotify at Scale

761M+	**Monthly Active Users** (Q1 2026, up 12% YoY)
293M+	**Premium Subscribers** (Q1 2026, up 9% YoY)
100M+	**Songs in Library** 6M+ Podcasts & Audiobooks

Source: Spotify Investor Relations, Spotify Newsroom

Spotify's Existing Discovery Engine

- AI DJ
- Smart Shuffle
- Daylist
- Discover Weekly
- Song Radio
- Personalized Mixes

Yet most listening still comes from familiar songs, repeat playlists, and known artists.

- Repeat playlists
- Familiar artists
- Previously discovered songs

The Core Gap

Spotify predicts what users liked yesterday—not what they want right now.

Spotify

Discovery Is Spotify's Biggest Growth Opportunity

Despite strong personalization, Spotify lags in meaningful discovery compared to competitors.

Music Platform Comparison

Platform	Strength	Discovery Experience
Spotify	Best UX, playlists, personalization	Safe but repetitive
YouTube Music	Excellent discovery via algorithm + videos	Best at surfacing new music
Apple Music	High-quality editorial curation	Good but less AI-driven
JioSaavn	Strong Indian catalog	Moderate discovery

Spotify optimizes for familiarity.

- Listening history
- Collaborative filtering
- Familiar artist similarity

This improves retention but often reduces novelty.

YouTube Music optimizes for curiosity.

- Shorts / reel behavior
- Video watch history
- Niche content discovery
- Creator ecosystem

Spotify's Problem

It's not lack of recommendations.

- Spotify optimizes for familiarity.
- YouTube optimizes for curiosity.

> Spotify says: "You liked this before."
> YouTube says: "You might find this interesting."
>
> That explains discovery differences.

Business Opportunity

Why this matters for Spotify Growth

Poor discovery leads to:

- Higher skip rates
- Shorter listening sessions
- Fallback to old playlists
- Reduced premium stickiness

Improve session-level discovery to increase engagement and retention.

Evidence from User Feedback

Common complaints across reviews:

> "Recommendations feel repetitive."

> "I discover better music on YouTube."

AI-Powered Review Discovery Engine

Scrape → Classify → Synthesize insights & answer questions across the full review corpus

3,171+ user reviews analyzed across 6 data sources

Why build this?

Manually analyzing thousands of reviews across multiple sources is slow, noisy, and inefficient. AI enables scalable pattern detection across unstructured feedback.

AI WORKFLOW (END-TO-END)

1 COLLECT REVIEWS

Ingest user feedback from 6 sources

- Google Play Store
- Apple App Store
- Reddit
- X (Twitter)
- Community Forums
- Social Media

Unified Corpus
3,171 records

2 AI CLASSIFICATION

LLM tags each review (batch size of 10)

- Theme
- Sentiment
- Key Quote
- Discovery Relevance

Pre-refine tagging
(batch size of 10)
temperature = 0.1

3 THEME EXTRACTION

AI extracts discovery themes and patterns

7 Discovery Themes Identified

- Recommendation Quality
- Repetition Fatigue
- Discovery Effort
- Algorithm Trust / Distrust
- Context Mismatch
- Social Identity
- Non-Discovery Related

4 RAG INSIGHT ENGINE

Retrieve, reason and answer with grounding

- Retrieve Top-15 Relevant Reviews (Cosine Similarity)
- Create Embeddings (all-MiniLM-L6-v2)
- Grounded LLM Answer (Cited Sources) Llama 3.3

5 OUTPUT

Actionable insights & interactive app

- Insights Report
- Theme Cards
- Q&A Chatbot
- Streamlit App

CAPABILITIES

The Engine Answers 6 Key Research Questions

- Why do users struggle to discover music?
- Common frustrations with recommendations?
- What listening behavior patterns emerge?
- Why do users repeat the same songs?
- Which segments face discovery issues?
- What unmet needs consistently appear?

Manual review of 3,000+ reviews is noisy and slow.

AI-powered engine for scalable pattern detection across unstructured feedback.

Transforms messy reviews into clear insights that drive better product decisions.

DEMO VIDEO https://drive.google.com/file/d/1yaV0FgtrhO0G-8CnwPS59hMTvZNjfZiZ/view?usp=share_link

LIVE APP https://spotify-review-ai.streamlit.app

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Key Insights from AI-Powered Review Analysis

Analysis of 3,171 reviews across 6 data sources revealed 4 major discovery gaps.

1 Spotify Over-Optimizes for Familiarity

Users repeatedly said Spotify recommends:

- Same artists
- Same genres
- Same songs
- Popularity-driven content

Evidence

- Recommendations feel repetitive and stale
- Algorithm prioritizes popular music over personal taste
- Users feel "stuck in a loop"

User Quote

"Spotify keeps playing the same songs over and over."

Instead of helping users discover fresh music.

2 Discovery Happens Outside Spotify

Users reported discovering music through:

- Instagram Reels
- YouTube
- Creators
- Friends

Then they return to Spotify to listen.

That means:

PM Insight

Spotify dominates listening, but not discovery.

3 Trust in Algorithm Is Declining

Users reported:

- Discover Weekly quality dropping
- Autoplay ruining flow
- AI recommendations feeling predictable

Quote

"One bad recommendation ruins the vibe."

4 Discovery Breaks When Context Changes ⭐

Users don't want generic recommendations.
They want music matching current session context.

Examples:

Driving · Gym · Focus · Relaxing · Night vibe

Problem:
Spotify recommends based on past behavior, not current intent.

Spotify understands taste, but not real-time intent.

Core Problem

Users do not struggle because Spotify lacks recommendations.
They struggle because Spotify cannot reliably balance:

Familiarity | Novelty | Session Intent | Vibe Continuity

"Spotify cannot balance historical familiarity with real-time session intent and song-level vibe similarity."

Spotify

User Research Validated Our AI Findings

Survey and interviews confirmed that Spotify's discovery problem is driven by context mismatch—not lack of content.

Quantitative Validation

Google Form Link

62 Survey Responses

- **58%** said Spotify recommendations feel **repetitive**
- **45.2%** said autoplay often repeats **similar songs**
- **43.5%** manually search for music **elsewhere**
- **30.6%** said recommendations don't match their **current mood**

Discovery dissatisfaction is common even among **active Spotify users.**

Qualitative Interviews

6 Semi-Structured Interviews

Interviewed:

Heavy Spotify Users	Active Music Explorers	Passive Playlist Listener
(3)	(2)	(1)

Common Behaviors Identified:

- Switch to YouTube for discovery
- Replay comfort playlists
- Search using moods instead of genres
- Want songs matching the **current moment**

Representative User Quotes



"I don't want similar artists. **I want songs that feel like this one."**

– Interviewee #2



Spotify knows my taste, **but not what I'm in the mood for."**

– Interviewee #4



"I usually discover songs on **Instagram or YouTube,** then come back to Spotify."

– Interviewee #5

Key Validation

AI analysis and primary research revealed the same pattern:

Users aren't asking for more recommendations—

they're asking for recommendations that understand their **current intent** and **emotional context.**

Spotify | Discovery Breaks Most for Context-Switching Heavy Listeners

1. Behavioral Segmentation Funnel

All Spotify Users

↓

Use music daily

↓

Multiple listening sessions per day

↓

Context-Switching Heavy Listeners

Behavioral segmentation is more predictive than demographic segmentation.

2. Selected User Segment

Primary User Segment

Heavy Spotify listeners who:

- listen 3–8+ times daily
- spend 2+ hours listening
- switch contexts frequently
- actively search for new music
- skip repetitive recommendations
- value song feel over artist familiarity

Common Contexts

- Commute
- Focus / Work
- Workout
- Night Unwind

Evidence from User Research

58%	45.2%	43.5%	30.6%
feel Spotify is repetitive	say autoplay repeats songs often	search outside Spotify for new music	say recommendations don't match their mood

These users consistently reported discovery challenges across multiple listening contexts.

3. Representative Persona



Rhea

- 26
- Product Designer
- Spotify Premium
- ~2 hours/day

Uses Spotify during

Commute | Work | Gym | Unwind

Frustrations

- Same songs again and again
- "Similar" recommendations feel wrong
- Has to search manually
- Switches to YouTube for discovery

Goal

Find unfamiliar songs that match the exact feel of the song I'm listening to, instantly.

JTBD

"I don't want the same artist again. I want songs that feel like this song."

Job To Be Done:

Help users instantly discover unfamiliar songs that match the exact vibe of what they're listening to right now.

Defining the Core Problem

The issue isn't a lack of content. It's a lack of understanding the listener's **current intent** and **emotional context**.

User Pain Points

Repetitive Recommendations
Daily Mixes and autoplay surface the same songs and artists.

High Discovery Effort
Users manually search YouTube, Instagram, or friends for new music.

Poor Mood Understanding
Recommendations don't match the user's current vibe or activity.

Discovery Fatigue
Finding fresh music takes too much time and effort.

Missed Hidden Gems
Users feel Spotify overlooks songs that actually fit their taste.

Root Cause: Why Existing Recommendations Fail

Today's Recommendation Engine Optimizes For:

Listening History
What users played before

↓

Artist & Song Similarity
What sounds similar to known music

↓

Popularity & Trends
What's popular among others

↓

Static Playlist Generation
Pre-built, not real-time

→ This misses what users actually need in the **moment**.

What Users Actually Need:

Current Song	Current Context	Current Mood	Song Feel Match	New Discovery
What they're listening to now	Where, what, what they're doing	How they feel right now	Songs that feel right for them	Unfamiliar songs that hit the right vibe

Key Insight

> **Spotify understands what users liked before, but not what they want to hear right now.**

Therefore...

Traditional recommendations systems optimize for:

- ✗ Listening history
- ✗ Artist similarity
- ✗ Popularity

But users expect recommendations based on:

- ✓ Current mood
- ✓ Current context
- ✓ Real-time intent

Core Problem:

Heavy Spotify listeners struggle to discover fresh music because the recommendation engine is optimized for **historical familiarity** rather than **real-time intent**.

Spotify Recommends Familiar Music, But Fails at Real Discovery

Problem Statement

Heavy Spotify listeners struggle to discover unfamiliar songs matching the exact feel of songs they love because Spotify prioritizes:

- **Familiarity**
- **Popularity**
- **Artist Similarity**

instead of **song-level vibe similarity**.



User Impact

Users experience:

- High skip rates
- Repetitive recommendations
- Manual effort to find fresh music
- Switching to YouTube for discovery



> "I love this song. Why can't Spotify help me find more songs that feel like this?"

Business Impact

This leads to:

- Lower discovery engagement
- Higher skip rates
- Reduced trust in recommendations
- Discovery happening outside Spotify



Business Risk:
Spotify becomes a **consumption platform**, not a **discovery platform**.

Goals

- Increase meaningful discovery
- Reduce repetition loops
- Enable discovery by musical feel

Non-Goals

- Rebuild Spotify recommendation engine
- Replace playlists / AI DJ
- Solve podcast recommendation

Spotify

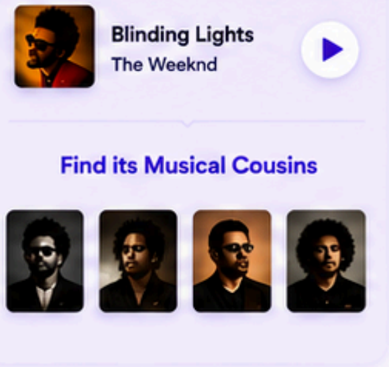
Meet VibePilot: Discovery by feel, not by artist

AI that understands your mood, context, and taste to help you **discover new music** you'll truly love.

Core Experience

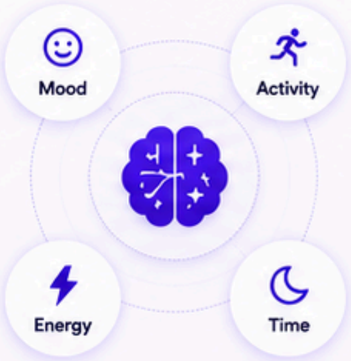
1 Find Cousins (Hero Feature)

Start with any song you already love. VibePilot finds its musical cousins—songs that feel similar, not just from the same artist.



2 Start From Vibe

Describe your mood, activity or energy in natural language to discover music that matches the moment.

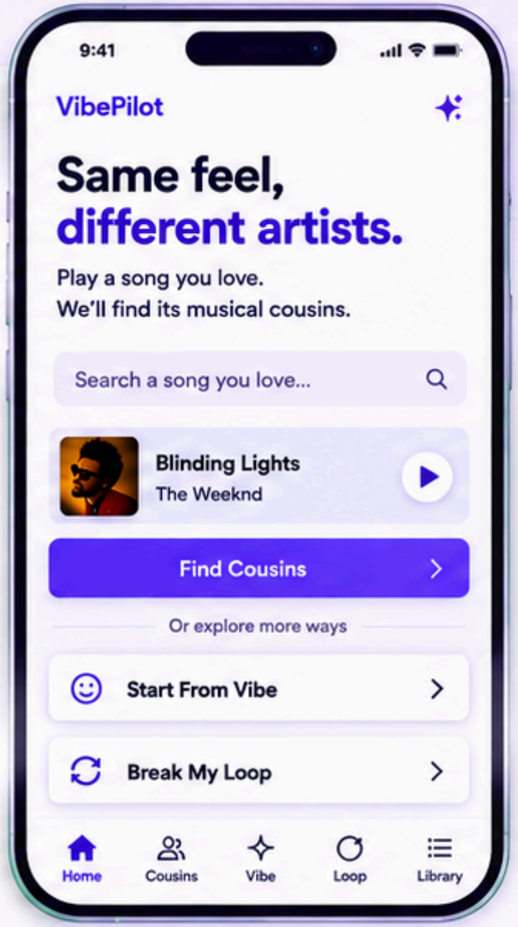


3 Break My Loop

Escape repetitive listening. Instead of replaying familiar artists, VibePilot surfaces fresh tracks that still feel familiar.

Break the loop.

- Step out of repetition
- Discover something fresh
- Stay in your vibe



Key Highlights

Find Cousins (Core)
Discover songs with the same feel, not the same artist.

Start From Vibe
Discover music from mood, activity or energy.

Break My Loop
Escape repetitive listening with intentional discovery.

AI Musical DNA
Matches tempo, energy, emotion and vibe—not only genres.

Personalized Discovery
Learns your evolving taste without trapping you inside recommendation loops.

> **Same feel. Different artists.**
> Play a song you love.
> We'll find its musical cousins.

Why VibePilot?

Solves the core discovery gap
Intent-driven, not history-driven.

Increases engagement
More saves, more shares, more retention.

Drives business impact
Higher listening time, premium stickiness, and user loyalty.

Feels magical
Personal, intuitive & delightful discovery.

Vibe Pilot https://vibepilot-ai.vercel.app | Demo Video https://drive.google.com/file/d/1gK4Pg8QW2koZPqMXUWqBlQsxAfiw5UjA/view?usp=share_link

Spotify

Measuring Success & Future Roadmap

Tracking real impact and building the future of music discovery.

NORTH STAR METRIC

Discovery Satisfaction Score (DSS)

Measures whether AI recommendations lead to meaningful music discovery.

DSS COMPONENTS

- **40%** **Songs Saved** — Users save songs they love
- **30%** **Playlist Adds** — Songs added to playlists
- **20%** **Song Replays** (within 7 days) — Songs re-listened to
- **10%** **Positive Feedback** — Thumbs up / positive rating

DSS gives us a holistic view of discovery quality, not just clicks or streams.

SUCCESS METRICS

Discovery Satisfaction Score (DSS)	Overall discovery quality	↑ +20% (vs baseline)
Songs Saved from AI Recommendations	Indicates love & value	↑ +25% (vs baseline)
Playlist Add Rate	Long-term value signal	↑ +15% (vs baseline)
Feature Adoption Rate	% of eligible users trying VibePilot	35%+ (in 90 days)
Weekly Returning Users (VibePilot)	Sustained engagement	40%+ (return rate)

These metrics help us measure meaningful impact, not just usage.

GUARDRAIL METRICS

Ensuring we improve discovery without hurting the core Spotify experience.

Skip Rate	Should decrease	↓
Session Duration	Should stay stable or increase	↔
30-Day User Retention	Should increase	↑
Recommendation Complaint Rate	Should decrease	↓
User Satisfaction (CSAT)	Should increase	↑

We grow responsibly while maintaining a great listening experience.

FUTURE ROADMAP

Continuously evolving to make discovery more personal, social & intuitive.

PHASE 1 (MVP)

- Conversational Discovery
- Discovery Dial
- AI Mood Understanding

Launched with core experience

PHASE 2

- **Collaborative Vibe Playlists** — Discover together with friends
- **Hyper-local Discovery** — More regional & culture-aware recs
- **Contextual Smart Mixes** — Activity, time & place aware mixes

PHASE 3

- **Personal Music Memory** — Learns your evolving taste over time
- **Voice Discovery** — "Catch that song."
- **AI DJ (Next Gen)** — Real-time, dynamic & interactive



Success is when users discover more music they love, more often.
VibePilot turns intent into discovery and discovery into loyalty.